MILITARY SPECIFICATION

PLUG, PROTECTIVE AND PRESERVATIVE, PLASTIC

(FOR INTERNAL THREADS)

1. SCOPE

1.1 Scope. - This specification covers a soft plastic plug for protecting internal female type threads.

1.2 Types and Sizes. - The plugs shall be of one type. Sizes shall be as specified in the invitation for bids, contract or order.

2. APPLICABLE DOCUMENTS

2.1 The following specifications of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

L-P-406 - Plastics, Organic; General Specifications, Test Methods.
P-S-661 - Solvent, Dry Cleaning
TT-C-595 - Colors, (for) Ready-Mixed Paints
LLL-B-631 - Boxes, Fiber, Corrugated (For Domestic Shipment)

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-M-10578 - Metal Conditioner and Rust Remover (Phosphoric Acid Base)

NAVY DEPARTMENT

General Specifications for Inspection of Material

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 - General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - The manufacturer is given wide latitude in the selection of materials, provided that the plastic plug conforms to the requirements of this specification. (See 6.2).

3.2 Design Features. - The plug shall be made of a soft, rubber-like thermo-plastic which will permit its insertion into female threads by means of a hand operated ram, and without screwing. Removal of the plug from the female threads shall also be possible by pulling on one edge, without unscrewing. The bottom end of the plug shall be closed with a thin diaphragm which it is practicable to remove from the top by suitable hand or power operated cutting out tools while the plug is in place. The threads of the plug shall be made to fit closely and firmly into the metallic female threads to assure adequate protection.

3.3 Manufacturing Process. - The plug shall be produced by injection molding. Split molds shall not be used.

3.4 Corrosion Protection. - The material of which the plug is made shall contain a rust inhibitor in the formulation (see 6.2) so that all surfaces of metallic female threads into which the plug is inserted will be protected against corrosion. After the test in 4.4.2 all surfaces of the metallic threads of the test specimens and other surfaces covered by the plug material shall be clean and free from rust.

3.5 Other Functional Requirements. - When tested in accordance with 4.4.3 and 4.4.4, the plug shall readily snap into the female threads, and provide a secure, intimate fit with the thread surfaces. The end diaphragm shall be easily removable with a mechanical cutout tool and the plug shall be wholly removable from the female thread opening.

3.6 Plasticizer and Rust Inhibitor Exudate. - During all of the tests of 4.4.2 to 4.4.4 inclusive, there shall be no noticeable exudate on the plug material itself or on adjacent surfaces of test specimens.

3.7 Color. - The color of the plug shall be equivalent to No. 2630 Specification TT-C-595 and shall be produced by a suitable pigment in the formulation.

3.8 Odors. - During all of the tests of 4.4.2 to 4.4.4 inclusive, the plugs shall not emit any offensive odors or vapors.

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4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection shall be in accordance with the provisions of MIL-STD-105.

4.2 Inspection. -

4.2.1 Government Inspection. - The Inspector shall, during the process of manufacture and assembly, perform, witness, or participate in such inspections and tests as are required by "Acceptance Inspection Instructions", and as otherwise considered necessary, to assure conformance with the requirements of the contract and all associated applicable documents. The "Acceptance Inspection Instructions" will normally be transmitted by the Procuring Activity by official correspondence at the time inspection cognizance is assigned.

4.2.2 Contractors Inspection. - The contractor shall determine, by whatever means or methods he desires and to the extent necessary, that all supplies and services and parts thereof offered for government acceptance or delivery conform with the requirements of the contract and all associated applicable documents.

4.2.3 Inspection Lot. - As applied to Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e., a collection of units of product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government Inspector and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.3 Size of Lot. - Each production lot shall consist of 20,000 plugs of the type and size being produced on a contract or order.

4.4 Tests. - The acceptance tests shall consist of all the tests described in 4.4.2 to 4.4.4 inclusive. Test samples shall consist of forty-five plugs randomly selected from each production lot.

4.4.1 Test Methods. -

4.4.2 Corrosion Test. - Thirty plugs shall be selected at random for tests.

4.4.2.1 Preparation of Test Specimens. - Metal specimens for test with the lot sample plugs shall be full size projectiles (empty) or specimens prepared from the same composition and having the same diameter, length of internal threads and machining tolerances. Immediately prior to the test, the threads of the metal specimens shall be cleaned as follows:

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Degrease with solvent conforming to Specification P-S-661 and remove any rust spots with rust remover conforming to MIL-M-10578. Rinse thoroughly in water, dry and then wash with solvent, Specification P-S-661. Follow this with two washes of absolute methyl alcohol. Dry in a silica gel desiccator for one hour.

4.4.2.2 Procedure. - Upon completion of the required time in the desiccator and immediately prior to the test the metal specimens shall be removed from the desiccator, using tongs or clean cotton gloves. At no time during the test shall the plastic plug or the metal specimen be touched with bare hands. The plastic plugs shall then be inserted into the metal test specimens by pushing on the bottom diaphragm with a round, flat-ended wooden dowel or metal ram approximately 1/4" smaller in diameter than the inside diameter of the plug. The specimens with plugs in place shall be immediately exposed to a relative humidity of 90-95% at a temperature of 110°F for 168 hours. Fifteen of the specimens shall be exposed to these conditions continuously. The other 15 shall be removed once a day during the first three days, placed in a refrigerator or cold box for 4 hours at 35 to 50 degrees F, then returned immediately to the original test conditions. At the completion of 168 hours the plugs shall be removed and the metal specimens carefully inspected for rust. The presence of rust on any of the metal specimens shall cause rejection of the lot of plugs.

4.4.3 Fitting Test. - Fifteen sample plugs shall be assembled into threads of the same size in a projectile or equivalent specimen (see 4.4.2.1) by pushing on the end diaphragm of the plug with a round, flat-ended wooden dowel or metal ram, approximately 1/4" smaller in diameter than the inside diameter of the plug, to determine whether the plug threads jump over the metal female threads easily, with no buckling or loose fitting of the closure after being seated. Acceptance will be in accordance with MIL-STD-105, Code ltr F, AQL 1%.

4.4.4 Diaphragm Removal. - After the fitting test of 4.4.3, the end diaphragms of the fifteen plugs shall be removed (while the plugs are in place in the thread opening) using a hand or power operated cutting tool. The portion of the diaphragm that is cut free shall follow out of the plug opening with the cutout tool and there shall be no burrs or rough edges left at the bottom edge of the plug which might interfere with projectile filling rams. Acceptance will be in accordance with MIL-STD-105, Code ltr F, AQL 1%.

4.4.5 Plug Removal. - Following the test of 4.4.4, the same plugs shall be tested by removing from the female thread openings by pulling on one edge with a pair of pliers, or with the fingers.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. - Plugs shall be packaged in fiberboard boxes, in accordance with LLL-B-631, Style RSC with B or C flutes.

5.2 Marking. - Marking of boxes shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. - The plastic plugs covered by this specification are intended for use as a protection for internal metal threads against corrosion during shipment and storage of unloaded projectiles, and as a shield against the collection of explosive in the threads during explosive filling of the projectiles at loading plants.

6.2 Formulation. - For information only the following formulation has been used for the manufacture of the item covered by this specification:

*Opalon 78263 resin	50.4%
Primary Plasticizer - Phthalate Type	25.0%
Other Plasticizer - Stabilizer	2.4%
Filler of the Atomite Type	20.7%
Stabilizer of the Barium-Calcium Type	1.5%
	100.0%

*Contains 5 parts or approximately 2 1/2% Rust Ban #626.

Patent Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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